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Lattice-based dynamic decentralized anonymous credential scheme supporting batch verification☆

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ABSTRACT

Anonymous credentials allow users to obtain credentials while protecting their privacy, which have significant application value in digital identity management systems. Anonymous credential schemes based on post-quantum assumptions mainly rely on a central issuing authority. Once the central server encounters single point of failure, it will cause the entire system to crash. The high computational complexity of credential verification algorithm severely restricts system scalability in high-concurrency identity management scenarios. More critically, existing schemes generally lack non-interactive batch verification algorithm, resulting in linear growth of verification time with the number of credentials when handling large-scale requests. Constructions relying on multiple fixed issuers to issue credentials in a decentralized manner have been proposed. Given that fixed issuers could not adapt to complex and changing network environments, we present a lattice-based decentralized anonymous credential system that supports dynamic changes in issuers to enhance system robustness. By introducing an innovative threshold signature, our system flexibly accommodates the joining or leaving of issuers for various reasons. Meanwhile, the system enables dynamic adjustments of thresholds without reset. By utilizing an efficient cryptographic accumulator, the system supports batch verification of credentials without revealing their contents. Experimental results demonstrate that our system outperforms previous systems in terms of performance. It is worth noting that our batch credential verification algorithm achieves constant time cost, independent of user's credential size.

1. Introduction

As network communication becomes increasingly advanced, information technology has become closely intertwined with people's lives. Digital identity which plays a significant role in user authentication is widely applied in various fields such as e-commerce and e-government [1]. The popularization of digital identity authentication technology has greatly facilitated people's daily lives [2]. For example, citizens can hold electronic identity cards to participate in online activities such as electronic voting. However, such electronic identity cards may expose users' personal information in practical use [3]. For instance, when users shop with credit cards associated with their electronic identity cards, service providers can easily link users' transaction records with their personal identities, thereby tracking their consumption behavior. It is crucial to protect privacy while performing identity authentication.

Anonymous credentials allow credential owners to disclose their identity information under pseudonyms, sticking up for their privacy in an anonymous manner [4]. Generally, in the anonymous credential system, users generate pseudonyms based on their identity information. Then, issuer identifies users through pseudonyms and issues credentials. Credential owners can use their credentials to prove to the verifier that they meet certain qualifications and gain corresponding services. Most of existing anonymous credential schemes are based on the RSA assumption [5] or the discrete logarithm assumption [6,7]. However, the quantum algorithms developed by Shor et al. in 1994 for integer factorization and discrete logarithm problems can quickly solve large prime factors, allowing quantum computers to break many cryptographic systems [8]. This means that the development of quantum computing threatens the security of almost all previous anonymous credential systems. Researchers have proposed lattice-based cryptography that remains secure against quantum attacks [9]. Therefore,schemes based on lattice have the security to withstand quantum attacks and are receiving increasing attention. After years of research, the current lattice-based cryptographic systems are mainly based on the following

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three hard problems: the LWE (Learning with Errors) problem [10], the SIS (Short Integer Solution) problem [9] and the NTRU (Number Theory Research Unit) problem [11]. Clearly, there is a need to design systems based on lattice problems to replace the previously insecure anonymous credential systems.

Traditional anonymous credential systems are prone to single point of failure, and researchers have proposed a large number of decentralized anonymous credential systems. Most decentralized anonymous credential systems are constructed based on the Shamir threshold scheme, requiring fixed issuers. However, due to various factors such as server failure or network instability, the issuers may leave, causing the system rewinding. In addition, most systems do not support batch verification of multiple credentials. Considering the limitations of anonymous credential systems mentioned above, we plan to design a lattice-based decentralized anonymous credential system. It is worth noting that we have abandoned the traditional Shamir Secret Sharing algorithm and innovatively proposed a new threshold signature scheme. This scheme can better cope with the dynamic changes of issuers in a decentralized environment, ensuring that the system can still operate stably when facing offline nodes or unstable networks. Besides, it supports dynamic changes in issuers and can adjust the threshold in a timely manner according to network environment, thereby enhancing the system flexibility and fault tolerance. The proposed system can resist quantum attacks and is suitable for the development of future computing technologies. In addition, we have introduced multi-credentials batch verification algorithm, effectively reducing communication and computing costs to improve efficiency.

The proposed design can provide users with efficient, secure, and privacy-preserving solutions in scenarios where sensitive data confidentiality needs to be ensured, such as cloud environments [12] and blockchain. For example, inspired by the use of revocable attribute-based encryption scheme to achieve fine-grained access control for user access permissions in cloud storage [13], we found a study that proposed a secure application management framework in the cloud. Combining anonymous credentials with this framework can effectively prevent identity fraud [14]. Even if attackers obtain some credential information, they cannot acquire user privacy. At the same time, this combination can achieve finer grained access control, where only users with specific attributes can access sensitive data or services.

1.1. Related work

Anonymous credentials: Anonymous credentials was proposed by Chaum in [15], which promoted research on privacy preserving. In 2001, Camenisch and Lysyanskaya provided the first concrete construction of an anonymous credential scheme based on the strong RSA assumption and the Diffie–Hellman assumption [16]. Brands and his team designed a method for credential issuance and verification using blind signatures and PKI certificates, and on this basis, they further developed the U-Prove system [17]. However, this system achieves unlinkability at the cost of non-repeatable use of credentials. Due to the flexibility of bilinear pairings, Camenisch and Lysyanskaya proposed a pairing-based signature scheme namely the CL signature building on the discrete logarithm assumption in [18] and further extended it to construct an anonymous credential system which adopts the "signature of a commitment" paradigm:allowing the user to commit to his private information and prove to the signer that his private information and commitment are consistent to obtain a signature. IBM implemented the CL signature and proposed the Identity Mixer system [19]. Identity Mixer supports anonymous identity verification and attribute transmission without collecting additional personal information, greatly protecting users' privacy. Most schemes satisfy unlinkability and unforgeability, which means that verifiers cannot link different credentials generated from the same user, and malicious users cannot forge credentials for valid use.

Decentralized anonymous credentials: The above schemes all rely on a central authority to issue credentials, which can easily lead to single point of failure. In addition, finding a single entity trusted by all participants is usually not desirable in practice. Garman et al. proposed the concept of Decentralized Anonymous Credentials (DAC) [20]. They advocate replacing the trusted credential issuer with Distributed Ledger Technology (DLT) such as blockchain, allowing users to obtain publicly verifiable credentials without the central issuer. Sonnino et al. proposed Coconut [21], which introduces multiple issuers in DAC to achieve threshold issuance credentials (i.e. at least t out of n honest issuers need to participate in issuing valid credentials). Coconut is based on the PS digital signature proposed by Pointcheval and Sanders [22]. Unlike PS signature, BBS+ signature has a constant public key [23]. Doerner et al. extended BBS+ signatures into a threshold anonymous credential scheme similar to Coconut [24] to realize issuing. Besides, their scheme supports tracing. Shi et al. proposed threshold attribute-based credentials [25], which have lower computational and communication complexity compared to Coconut, and support tracing and revocation. Most of these schemes rely on the Shamir's secret sharing protocol which requires issuers to be fixed when establishing the system. Therefore, after system initialization, not only does changing issuers require system rewinding, but the threshold cannot be dynamically adjusted. In many scenarios, the system has low availability.

Lattice-based anonymous credentials: Although various efficient constructions for anonymous credentials have been proposed, the accomplishments in the post-quantum are extremely limited. Notably, up to now, lattice-based anonymous credential schemes have gradually attracted the attention of researchers. Lai et al. aiming to promote the development of anonymous credential systems in post-quantum, proposed lattice-based commit-transferrable signatures (CTS) in [26]. Combining CTS with a multi-theorem straight-line extractable non-interactive zeroknowledge proof of knowledge (NIZKPoK), they designed an anonymous credential system. Due to the high computational cost of system initialization and key generation, as well as excessive reliance on a single issuer, the cost of system rewinding is too high in the event of issuer server failure. In practical applications, the system needs to have sufficient fault tolerance to handle single point of failure, which is too difficult to implement. Jeudy et al. noticed that the signature scheme called signatures with efficient protocols proposed by Libert et al. [27] yields very large signatures and proofs. They proposed a similar scheme in [28] and extended it to a lattice-based anonymous credential system, while the computational cost of generating credentials and proofs is closely related to the number of attributes. When there are a large number of attributes, generating a single credential requires significant computational resources, which may result in a substantial reduction in system efficiency. Bootle et al. presented a lattice-based framework for constructing practical anonymous credential schemes [29]. This framework has strong scalability and can be flexibly applied to various scenarios. However, its verification process is relatively complex, especially when it involves the verification of multiple credentials, resulting in low efficiency, which to some extent limits its application in large-scale scenarios. All of these schemes rely on a central authority to issue credentials, which makes the system less robust. Once the central authority malfunctions or comes under attack, it will cause system failure or interruption of credential issuance.

1.2. Our contribution

We propose a decentralized anonymous credential system that can resist quantum attacks. The system allows issuers to issue credentials in a threshold manner and dynamically change status while supporting threshold adjustments. In addition, the system is capable of batch verifying multiple credentials.

Our contributions are listed below:

1. We propose an innovative threshold signature scheme that supports dynamic adjustment of issuers. We have noticed that most of the current threshold signature schemes are based on the Shamir's protocol, which requires issuers to be fixed once initialized, resulting in a lack of flexibility in existing threshold anonymous credential systems. The system built on our scheme supports issuers to frequently join or leave for various reasons without the need for rewinding.
2. We apply cryptographic accumulators to verification, transforming the credential verification into membership proof, which enables the verifier to perform batch verification when multiple credentials need to be verified, thereby improving verification efficiency. In addition, our system ensures that verifier do not know specific contents of user's credentials, but only the quantity. It is particularly applicable when users are involved in various activities organized by distinct institutions during the same process.
3. We design an anonymous credential system that allows users to apply for credentials anonymously in a semi-honest decentralized environment. Our system meets the requirements of unforgeability, blindness, and unlinkability.
4. We present how to instantiate our system based on lattice assumptions, which provides security against quantum attacks.

2. Preliminaries

2.1. Notation

The notations utilized in this paper are compiled and presented in Table 1.

2.2. Lattices

In the context of cryptography, discrete additive subgroups in $\mathbb{R}^m$ can be considered as lattices. Typically,if we have a set of vectors consisting of m linearly independent n-dimensional vectors, that is $(\mathbf{b}_1, \ldots, \mathbf{b}_m)$. We can then construct a lattice Λ using $(\mathbf{b}_1, \ldots, \mathbf{b}_m)$ as follows:

$$\Lambda = \mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_m) = \left\{ \sum_{i=1}^{m} z_i b_i \ : \ z_i \in \mathbb{Z} \right\}.$$

The lattice Λ is referred to as full-rank if n=m.

Definition 2.1. Let n, m, q be positive integers with $q \geq 2$. For $A \in \mathbb{Z}_q^{n\times m}$ and $u \in \mathbb{Z}_q^n$ in the subgroup of $\mathbb{Z}_q^n$ generated by the columns of A, define:

$$\Lambda^{\perp}(\mathbf{A}) = \{\mathbf{e} \in \mathbb{Z}^m \ : \ \mathbf{Ae} = \mathbf{0} \mod q\}, \Lambda^{\mathbf{u}}(\mathbf{A}) = \{\mathbf{e} \in \mathbb{Z}^m \ : \ \mathbf{Ae} = \mathbf{u} \mod q\}.$$

For all $\mathbf{c} \in \mathbb{R}^n$ and any $\sigma > 0$, the Gaussian function centered at $\mathbf{c}$ with standard deviation parameter σ is defined as:

$$\forall \mathbf{x} \in \mathbb{R}^n, \rho_{\sigma,\mathbf{c}}(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x}-\mathbf{c}\|^2}{2\sigma^2}\right).$$

For all $\mathbf{c} \in \mathbb{R}^n$, any $\sigma > 0$ and an n-dimensional lattice Λ, the discrete Gaussian distribution over Λ centered at $\mathbf{c}$ with standard deviation parameter σ is defined as:

$$\forall \mathbf{x} \in \Lambda, D_{\Lambda,\sigma,\mathbf{c}} = \frac{\rho_{\sigma,\mathbf{c}}(\mathbf{x})}{\rho_{\sigma,\mathbf{c}}(\Lambda)},$$

where $\rho_{\sigma,\mathbf{c}}(\Lambda) = \sum_{\mathbf{x}\in\Lambda} \rho_{\sigma,\mathbf{c}}(\mathbf{x})$.

2.3. Lattice trapdoors

We will employ a pair of algorithms for two purposes: one for generating a random lattice with an embedded trapdoor and another for sampling short vectors within a specific lattice coset. The algorithm for generating lattices with trapdoors is built upon the foundations laid by the research presented in [30,31], whereas the algorithm for sampling short vectors is informed by the insights described in [32,33].

Table 1
Notations used.

Notation	Description
$[\mathbf{A}\|\mathbf{B}]$	The horizontal concatenation of matrices $\mathbf{A}$ and $\mathbf{B}$
$[m]$	The set $\{1, 2, \ldots, m\}$
$\|I\|$	The cardinality of set I
$\mathbb{Z}_q$	The cyclic group with addition modulo q
$\mathbf{v}[i]$	The ith element of $\mathbf{v}$
$\|\cdot\|$	The Euclidean (ℓ_2) norm
$\|\cdot\|_\infty$	The infinity (ℓ_∞) norm
$\|\tilde{\mathbf{A}}\|$	The Gram–Schmidt norm of $\mathbf{A}$
$\mathbf{A}^T$	Transpose of matrix $\mathbf{A}$
$poly(n)$	Polynomial function of n
$negl(n)$	Negligible function of n
ϵ	Negligible probability
$\{0,1\}^*$	Binary string of arbitrary length

Lemma 2.2. *Let n, m, q be positive integers with $q \geq 2$ and $m = O(n \log q)$, there exists a Probabilistic Polynomial Time algorithm,denoted as TrapGen(1^n, 1^m, q): given n, q as well as m, it outputs a trapdoor pair $(\mathbf{A}, \mathbf{T}) \in \mathbb{Z}_q^{n\times m} \times \mathbb{Z}_q^{m\times m}$ such that the matrix $\mathbf{A}$ is within $2^{-\Omega(n)}$ statistical distance to uniform in $\mathbb{Z}_q^{n\times m}$ and $\mathbf{T}$ is a basis for the left null space $\Lambda_q^{\perp}(\mathbf{A})$.*

Lemma 2.3. *Let n, m, q be positive integers with $q \geq 2$ and $m = O(n \log q)$, there exists a Probabilistic Polynomial Time algorithm,denoted as SamplePre($\mathbf{A}$, $\mathbf{T}$, $\mathbf{v}$, σ):given a matirx $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ with a short basis $\mathbf{T} \in \mathbb{Z}_q^{m\times m}$, parameter $\sigma \geq \omega(\sqrt{\log n}) \cdot \|\tilde{\mathbf{T}}\|$ and a vector $\mathbf{v} \in \mathbb{Z}_q^n$, it outputs a vector $\mathbf{e} \in \mathbb{Z}^m$ for a disribution statistically close to $D_{\Lambda^{\mathbf{u}}(\mathbf{A}),\sigma}$.*

2.4. Hardness assumptions

We will utilize the Learning With Errors (LWE) problem and Short Integer Solution (SIS) problem.

Lemma 2.4 (*Learning With Errors*)**.** *Let q, n, m be parameters that depend on a security parameter λ. Let $\mathbf{s} \in \mathbb{Z}_q^n$ be a secret vector. Define a distribution $D_{\mathbf{s},\chi}$ over $\mathbb{Z}_q^n \times \mathbb{Z}_q$ as follows: for each sample in the distribution, a vector $\mathbf{a}$ is chosen uniformly at random from $\mathbb{Z}_q^n$ and an error term e is sampled from a probability distribution χ. The pair $(\mathbf{a}, \mathbf{a}^\top \cdot \mathbf{s} + e)$ is then output. The LWE problem is to distinguish: given $m = poly(n)$ samples, whether these samples are from the distribution $D_{\mathbf{s},\chi}$ or $\mathbb{Z}_q^n \times \mathbb{Z}_q$.*

Lemma 2.5 (*Short Integer Solution*)**.** *Let q, n, m, β be parameters that depend on a security parameter λ. An instance of the SIS problem is defined by a matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$. The solution is finding a nonzero vector $\mathbf{v} \in \mathbb{Z}^m$ that satisfies two conditions: if $\beta = poly(n)$ then $\|\mathbf{v}\| < \beta$ and $\mathbf{A} \cdot \mathbf{v} = \mathbf{0} \pmod q$.*

2.5. Non-interactive zero-knowledge proof

Zero-Knowledge proof is a cryptographic method that allows one party to prove to another that a statement is true, without providing any private information [34].

Definition 2.6. A Non-Interactive Zero-Knowledge Proof(NIZK) consists of three algorithms that adhere to the following syntax:

- **Gen**(1^λ): On input a security parameter λ, the Gen algorithm outputs the parameter pp.
- **Prove**(pp, x, ω): On input pp, a statement x and the corresponding witness ω, the prover generates a proof π.
- **Verify**(pp, π, x): The Verifier checks the validity of statement x through proof π. If valid, the Verifier can be assured of the correctness of statement x without knowing details.

The NIZK must satisfy the following key properties:

- *Completeness*: If a statement is true, the prover can convince the verifier to accept it. This property ensures that a system works correctly in the absence of malicious behavior. If a system lacks completeness, the true statement might not be accepted by the verifier, which would greatly reduce practicality of the system.
- *Soundness*: If a statement is false, the prover convinces the verifier that it is computationally infeasible, even with infinite computing power. This property ensures the system cannot be exploited by fraudulent actions. If a system lacks soundness, malicious provers might deceive the verifier by fabricating proof, undermining security of the system.
- *Zero-Knowledge*: Zero-knowledge is the core characteristic ensuring that the verifier, during the verification process, cannot gain any knowledge about the prover's private information. This means that even if the verifier verifies the same statement multiple times, he cannot extract any additional information.
- *Non-Interactivity*: The proof process only requires a single round of communication, where the prover sends the proof to the verifier, who then independently verifies the proof's validity without further interaction.

2.6. Dynamic accumulator scheme

Cryptographic accumulators offer a method for generating a compact and definitive commitment to a collection of values. Our system leverages the dynamic accumulator scheme, which was originally proposed by Derler [35] and subsequently improved by other researchers [36].

Definition 2.7. A lattice-based cryptographic accumulator scheme includes the following algorithms:

- **ACC.KeyGen**(1^λ): Obtain $\mathbf{A}', \mathbf{T}' \leftarrow \mathrm{TrapGen}(1^a, 1^b, q)$. Next, sample a random matrix $\mathbf{U} = [\mathbf{U}_1 | \cdots | \mathbf{U}_M] \in \mathbb{Z}_q^{a\times(c\cdot M)}$, where $\mathbf{U}_i \in \mathbb{Z}_q^{a\times c}$. Then, for each $i \in [M]$, sample $\mathbf{S}_i \leftarrow \mathrm{SamplePre}(\mathbf{A}', \mathbf{T}', \mathbf{U}_i, \alpha)$ which means $\mathbf{A}'\mathbf{S}_i = \mathbf{U}_i$. Output the public key $Apk = (\mathbf{U}, \mathbf{A}')$ and the secret key $Ask = \mathbf{S} = (\mathbf{S}_i)_{i\in[M]} \in \mathbb{Z}^{b\times(c\cdot M)}$.
- **ACC.Eval**$(Apk, X \subseteq [M])$: Construct the vector $\mathbf{z} = [\mathbf{z}_1^T | \cdots | \mathbf{z}_M^T]^T$, where each $\mathbf{z}_i$ is a c-dimensional nonzero vector with all terms are 1 if $i \in X$, otherwise, $\mathbf{z}_i = \mathbf{0}^c$. Next, sample a random vector $\mathbf{r} \leftarrow D_{\mathbb{Z}_q^b,\alpha}$. Then, compute and output the accumulator value $\mathbf{v} = \mathbf{U}\mathbf{z} + \mathbf{A}'\mathbf{r} \in \mathbb{Z}_q^a$.
- **ACC.WitGen**$(Ask, \mathbf{v}, \mathbf{U}, i \in X)$: Output the witness for $i \in X$

  $$\mathbf{w} = \sum_{j\in X, j\neq i} \mathbf{S}_j\mathbf{z}_j + \mathbf{r} \in \mathbb{Z}_q^b,$$

  where $\mathbf{z}_j$ is calculated from X.
- **ACC.Verify**$(Apk, \mathbf{v}, j, \mathbf{w})$: Accept if $\|\mathbf{w}\|_\infty \leq \gamma$ (or $\|\mathbf{w}\| \leqslant \gamma_1$) and $\mathbf{v} = \mathbf{A}'\mathbf{w} + \mathbf{U}\cdot\mathbf{y}$, where $\gamma = O(\alpha\sqrt{cM}) < q$, $\gamma_1 = O(\alpha(b + cM)) < q$, and $\mathbf{y} = [0^c | \ldots | \mathbf{y}_j^T | \ldots | 0^c]^T$, $\mathbf{y}_j$ is a c-dimensional nonzero vector that all terms are 1.

3. Our scheme

3.1. System model

Fig. 1 is a schematic diagram of the proposed system, consisting of five basic entities: the Trusted Authority, the Issuers, the Adjudication Committee, the User and the Verifier.

- Trusted Authority: The Trusted Authority is not only responsible for carrying out the system initialization, ensuring that all parameters are correctly configured according to established standards, but also tasked with generating keys for certain participants within the system, including the Issuers and the Adjudication Committee.

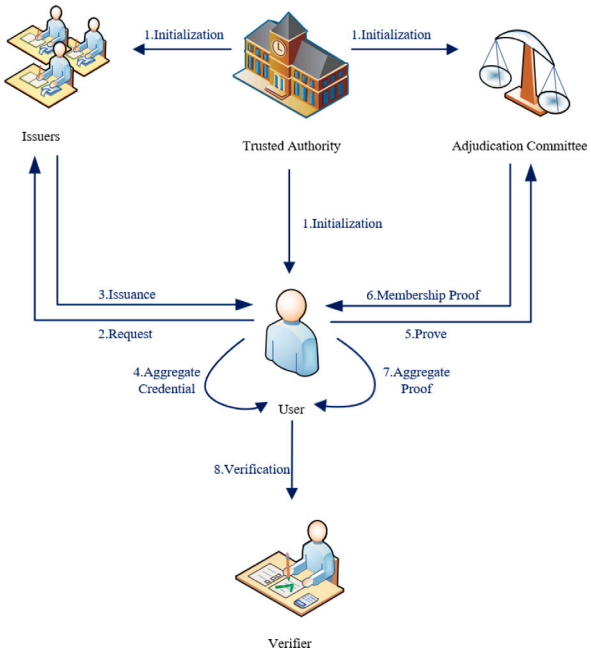


Fig. 1. System overview.

- Issuers: The Issuers are responsible for rigorously verifying the pseudonym of User to ensure legality of the credential. Once the pseudonym verification is successful, each issuer will distribute credential share to the User. In our scheme, there are m issuers in all and the initial threshold is t. In practical application, the Issuers can freely join or leave the system. Meanwhile, the threshold is also dynamically adjustable.
- Adjudication Committee: Carefully review each submitted credential to ensure its validity while being curious about User's private information.After confirming that the credentials are correct, the Adjudication Committee will generate witness of member relationship for each credential, which will clearly demonstrate the connection between credentials and the owner.
- User: As the applicant for credentials, the User generates pseudonyms to protect true identity information. After obtaining the credential shares, the User can aggregate them to obtain a single full credential which is used to access service. As the owner of multiple credentials, the user can aggregate membership witnesses corresponding to valid credentials which have been verified by the Adjudication Committee.
- Verifier: In the batch verification phase, the Verifier will honestly check the aggregated membership witness submitted by the User to verify User's credential set. However, the Verifier may be curious about User's credential set.

3.2. Scheme definition

We formalize our scheme into the following algorithms:

- **Setup**(1^λ): Given security parameter λ, Trusted Authority runs **ACC.KeyGen** to get Apk and Ask and outputs the system parameter pp.
- **PdnGen**(pp, μ): To obtain a credential share, User inputs his private information μ and pp. Outputs the pseudonym pdn and a Non-Interactive Zero-Knowledge Proof Θ for the pdn.
- **KeyGen**(pp, m, t): Given pp, the total number of Issuers m and a threshold t.Trusted Authority computes secret key shares $(sk_1, \ldots, sk_n)$ and corresponding public key shares $(pk_1, \ldots, pk_n)$ for Issuers. sk_i and pk_i will be sent to $Issuer_i$ through a secure channel.

- **Issue**(pdn, Θ, pp, sk_i): Given pp, pdn and the corresponding Non-Interactive Zero-Knowledge Proof Θ, $Issuer_i$ verifies the validity of pdn according to Θ, then issues a credential share $cred_i$ to User.
- **AggreCred**($\{cred_i\}_{i\in I}$): After collecting $|I|(|I| \geq t_I)$ credential shares, User aggregates these credential shares into a single credential $Cred$, stores the indices of valid credential issuers as the bit vector **b**.
- **TransCred**($Cred$, **b**, $\{pk_i\}_{i\in[m]}$, pdn): User computes the aggregated public key pk of valid issuers according to **b**,transforms his pdn and $Cred$ into pdn' and $Cred'$.
- **VerCred**($Cred'$, **b**, $\{pk_i\}_{i\in[m]}$, pk, pdn'): According to **b** and $\{pk_i\}_{i\in m}$, Adjudication Committee recomputes pk and determines if they are equal. Then Adjudication Committee parses pdn' and $Cred'$, checks their correctness. If all the statements hold, Adjudication Committee accepts the credential, then runs **ACC.WitGen** and **ACC.Eval** to generate a membership witness **w** for $Cred'$ and update the accumulator value **v**.
- **AggreWit**($\mathbf{w}_1, \ldots, \mathbf{w}_N$): If User has multiple valid credentials for accessing distinct services such as qualifications to take part in different activities, User has the capability to aggregate corresponding membership witnesses into a single witness $\mathbf{w}'$.
- **BatchVer**($\mathbf{w}'$, **v**, pp): If User owns a set of valid credentials,given aggregated membership witnesses $\mathbf{w}'$ and the accumulator value **v**,Verifier validates their correctness.

3.3. Security definition

3.3.1. Threat model

Our threat model is based on the following assumptions about participating entities:

- Trusted Authority is a trusted entity responsible for system initialization and key generation. It is considered reliable, and all parties involved rely on Trusted Authority to obtain necessary information. We assume that there is no collusion risk between Trusted Authority and users.
- Issuers are regarded as entirely trustworthy. However, their online status may experience fluctuations due to various unpredictable factors, such as network instability or server failure.
- Adjudication Committee is assumed to be honest but curious. It will faithfully perform its tasks as scheduled, yet it may have an interest in users' hidden information. On the other hand, Adjudication Committee might collude with users to extract additional information.
- User is considered to be dishonest and have a tendency towards collusion. It may attempt to forge credentials to gain unauthorized access or manipulate information, posing significant security risks and potentially compromising system integrity and confidentiality.
- Verifier is assumed to be honest but curious. It will faithfully perform its tasks as scheduled, yet it may have an interest in user's credential set.

3.3.2. Security model

Next, we will define three games between a challenger C and a Probabilistic Polynomial Time adversary $\mathcal{A}$ to explain the security of our scheme, which is similar to [6,37,38]. $\mathcal{A}$ is granted access to certain oracles. These oracles are programs that simulate components of our scheme, allowing $\mathcal{A}$ to interact with the scheme in specific ways.

- Unforgeability: The property of unforgeability ensures that it is infeasible for a malicious user to create a counterfeit credential using fewer than t credential shares, or to deceive the verification process.Our scheme achieves Existential Unforgeability under Chosen-Message Attacks (EU-CMA) security, which means that even if an adversary $\mathcal{A}$ can make queries to various oracles, they cannot produce a fraudulent credential that would be accepted by the system.

$\mathrm{Exp}_{\mathcal{A}}^{\mathrm{EU-CMA}}(1^\lambda)$:

1. $pp \leftarrow \mathbf{Setup}(1^\lambda)$;
2. $\{st, sk_i, pk_i\} \leftarrow \mathbf{KGen}(pp, m, t)$;
3. $\mathcal{O} \leftarrow \{\mathcal{O}_{PdnGen}, \mathcal{O}_{Issue}, \mathcal{O}_{AggreCred}\}$;
4. $Cred^* \leftarrow \mathcal{A}^{\mathcal{O}}(st)$;
5. $\mathbf{v}, \mathbf{w} \leftarrow \mathbf{VerCred}(Cred^*, \mathbf{b}, \{pk_i\}_{i\in m}, pk, pdn)$.

Fig. 2. Game $\mathrm{Exp}^{\mathrm{EU-CMA}}$.

$\mathrm{Exp}_{\mathcal{A}}^{\mathrm{Blind}}(1^\lambda)$:

1. $pp \leftarrow \mathbf{Setup}(1^\lambda)$;
2. $\{st, pk_i\} \leftarrow \langle \mathcal{A}(\mathbf{KGen}, pp)\rangle$;
3. $\mathcal{O} \leftarrow \{\mathcal{O}_{Issue}, \mathcal{O}_{Blind}\}$;
4. $b \leftarrow \{0, 1\}$;
5. $b^* \leftarrow \mathcal{A}^{\mathcal{O}}(st)$;
6. $b^* = b$.

Fig. 3. Game $\mathrm{Exp}^{\mathrm{Blind}}$.

Definition 3.1 (*Unforgeability*)**.** The following game $\mathrm{Exp}^{\mathrm{EU-CMA}}$ defines the unforgeability of our scheme in Fig. 2. Our scheme is unforgeable if for any $\mathcal{A}$ the game outputs 1 that possesses advantage:

$$\mathrm{Adv}_{\mathcal{A}}^{\mathrm{EU-CMA}} = \left|\Pr[\mathrm{Exp}_{\mathcal{A}}^{\mathrm{EU-CMA}}(1^\lambda)] = 1\right| \leq \mathrm{negl}(\lambda)$$

- Blindness: The property of blindness serves as a shield for User's private details, safeguarding them from the scrutiny of the Issuers.This ensures that User's identity and associated sensitive data remain confidential throughout the credential issuance process, preventing any unauthorized access or exposure.

Definition 3.2 (*Blindness*)**.** The following game $\mathrm{Exp}^{\mathrm{Blind}}$ defines the blindness of our scheme in Fig. 3. Our scheme is blind if for any $\mathcal{A}$ the game outputs 1 that possesses advantage:

$$\mathrm{Adv}_{\mathcal{A}}^{\mathrm{Blind}} = \left|\Pr[\mathrm{Exp}_{\mathcal{A}}^{\mathrm{Blind}}(1^\lambda)] - \frac{1}{2}\right| \leq \mathrm{negl}(\lambda)$$

- Unlinkability: The property of unlinkability ensures that different credentials issued to an individual cannot be linked to the same owner.This makes certain that malicious verifier cannot trace the User's behavior.

Definition 3.3 (*Unlinkability*)**.** The game $\mathrm{Exp}^{\mathrm{Unlink}}$ defines the unlinkability of our scheme in Fig. 4. Our scheme is unlinkable if for any $\mathcal{A}$ the game outputs 1 that possesses advantage:

$$\mathrm{Adv}_{\mathcal{A}}^{\mathrm{Unlink}} = \left|\Pr[\mathrm{Exp}_{\mathcal{A}}^{\mathrm{Unlink}}(1^\lambda)] - \tfrac{1}{2}\right| \leq \mathrm{negl}(\lambda)$$

3.4. Construction

- **Setup**(1^λ)
  Given security parameter λ,Trusted Authority carries out the following steps:

  1. Set the parameters n, k, ℓ, m(ℓ is dividable by m).
  2. Set $a = a(\lambda)$, $M \in \mathbb{M} > 2$, $q = \mathrm{poly}(a) > M$, $b = O(a \log q)$, $c = \lceil \log_2 M \rceil$.

$\mathrm{Exp}_{\mathcal{A}}^{\mathrm{Unlink}}(1^{\lambda})$:

1. $pp \leftarrow \mathbf{Setup}(1^{\lambda})$;
2. $\{st, pk_i\} \leftarrow \langle \mathcal{A}(\mathbf{KGen}, pp)\rangle$;
3. $\{0,1\} \leftarrow b$;
4. $(\mathbf{w}_0', \mathbf{w}_1')_b \leftarrow \mathbf{AggreWit}(\{(\mathbf{w}_1, \ldots, \mathbf{w}_N)_0, (\mathbf{w}_1, \ldots, \mathbf{w}_N)_1\}_b)$;
5. $\mathcal{O} \leftarrow \{\mathcal{O}_{Issue}, \mathcal{O}_{Unlink}\}$;
6. $b^* \leftarrow \mathcal{A}^{\mathcal{O}}(st)$;
7. $b^* = b$.

Fig. 4. Game $\mathrm{Exp}^{\mathrm{Unlink}}$.

3. Set a gaussian parameter α.
4. Sample $\mathbf{A} = \begin{bmatrix} \mathbf{I}_n, \mathbf{A}_1 \\ \mathbf{0}^{\ell\times n}, \mathbf{A}_2 \end{bmatrix}$, where $[\mathbf{I}_n, \mathbf{A}_1] \in \mathbb{Z}_q^{n\times k}$, $[\mathbf{0}^{\ell\times n}, \mathbf{A}_2] \in \mathbb{Z}_q^{\ell\times k}$, $\mathbf{A}_1 \in \mathbb{Z}_q^{n\times(k-n)}$, $\mathbf{A}_2 \in \mathbb{Z}_q^{\ell\times(k-n)}$.
5. Obtain $\mathbf{A}', \mathbf{T}' \leftarrow \mathrm{TrapGen}(1^a, 1^b, q)$.
6. Sample a random matrix $\mathbf{U} = [\mathbf{U}_1 | \cdots | \mathbf{U}_M] \in \mathbb{Z}_q^{a\times(c\cdot M)}$, where $\mathbf{U}_i \in \mathbb{Z}_q^{a\times c}$.
7. For each $i \in [M]$, sample $\mathbf{S}_i \leftarrow \mathrm{SamplePre}(\mathbf{A}', \mathbf{T}', \mathbf{U}_i, \alpha)$.
8. Let the accumulator public key $Apk = (\mathbf{U}, \mathbf{A}')$ and the accumulator secret key $Ask = \mathbf{S} = (\mathbf{S}_i)_{i\in[M]} \in \mathbb{Z}^{b\times(c\cdot M)}$.
9. Set $\gamma = O(\alpha\sqrt{cM}) < q$, $\gamma_1 = O(\alpha(b + cM)) < q$.
10. Define a secure hash function $H : \mathbb{Z}_q^{2k+2\ell\cdot m-n} \to \mathbb{Z}_q^*$.
11. Sample a random matrix $\mathbf{D} \xleftarrow{\$} \mathbb{Z}_q^{\ell\times k}$.
12. Let $\mathbf{G} = \mathbf{I}_\ell \otimes \mathbf{g}^T \in \mathbb{Z}_q^{\ell\times(\ell\cdot m)}$, where $\mathbf{I}_\ell \in \mathbb{Z}_q^{\ell\times\ell}$ is the identity matrix and $\mathbf{g}^T = (1, \delta, \ldots, \delta^{m-1})$, $\delta = \lfloor q^{1/m} \rfloor$.
13. Output the system parameter $pp = (\mathbf{A}, \mathbf{D}, Apk, H, \mathbf{G}, \alpha)$, send Ask to Adjudication Committee.

- **PdnGen**(pp, μ)
  To obtain a credential share, User inputs his private information μ and pp. Then, the algorithm runs these steps:

1. User sample $\mathbf{R} \xleftarrow{\$} \mathbb{Z}_q^{(k-n)\times(\ell\cdot m)}$.
2. Compute $pdn = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{bmatrix} \cdot \mathbf{R} + \begin{bmatrix} \mathbf{0}^{n\times(\ell\cdot m)} \\ \mu \cdot \mathbf{G} \end{bmatrix} \in \mathbb{Z}_q^{(n+\ell)\times(\ell\cdot m)}$.
3. Compute the proof Θ:

$$NIZK\left\{(\mu, \mathbf{R}) : pdn = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{bmatrix} \cdot \mathbf{R} + \begin{bmatrix} \mathbf{0}^{n\times(\ell\cdot m)} \\ \mu \cdot \mathbf{G} \end{bmatrix} \in \mathbb{Z}_q^{(n+\ell)\times(\ell\cdot m)}\right\}.$$

4. Output the pseudonym pdn and corresponding Non-Interactive Zero-Knowledge Proof Θ.

- **KeyGen**(pp, m, t)
  Given pp, the total number of Issuers m and a threshold t, Trusted Authority outputs sk_i and corresponding pk_i for $Issuer_i$:

1. Trusted Authority select a random matrix $\mathbf{T} = [\mathbf{T}_1 | \cdots | \mathbf{T}_m] \in \mathbb{Z}_q^{k\times(\ell\cdot m)}$, $\mathbf{T}_i' = [0^\ell | \cdots | \mathbf{T}_i | \cdots | 0^\ell] \in \mathbb{Z}_q^{k\times(\ell\cdot m)}$.
2. Compute $\mathbf{A}_{0,i} = \mathbf{D} \cdot \mathbf{T}_i' + \mathbf{G} \in \mathbb{Z}_q^{\ell\times(\ell\cdot m)}$.
3. Sample a random matrix $\mathbf{B} \xleftarrow{\$} \mathbb{Z}_q^{\ell\times(\ell\cdot m)}$ and a non-zero vector $\mathbf{u} \xleftarrow{\$} \mathbb{Z}_q^{\ell}$.
4. Output $pk_i = (\mathbf{A}_{0,i}, \mathbf{B}, \mathbf{u})$, $sk_i = \mathbf{T}_i'$.

- **Issue**$(pdn, \Theta, pp, pk_i, sk_i)$: To issue a credential share, $Issuer_i$ inputs pdn, Θ, pp, sk_i and executes this algorithm through the following steps:

1. $Issuer_i$ verifies the validity of pdn according to Θ.
2. Parse $pdn = \begin{bmatrix} \mathbf{C}_1 \\ \mathbf{C}_2 \end{bmatrix}$, where $\mathbf{C}_1 \in \mathbb{Z}_q^{n\times(\ell\cdot m)}$, $\mathbf{C}_2 \in \mathbb{Z}_q^{\ell\times(\ell\cdot m)}$.
3. Set $\mathbf{F}_i = [[\mathbf{D} | \mathbf{A}_{0,i}] | \mathbf{B} + \mathbf{C}_2 | \mathbf{A}_2]$.
4. Sample $cred_i = \begin{bmatrix} \mathbf{s}_{1,i} \\ \mathbf{s}_{2,i} \\ \mathbf{s}_{3,i} \end{bmatrix} \leftarrow \mathrm{SamplePre}(\mathbf{F}_i, \mathbf{T}_i', \mathbf{u}, \alpha)$, where $\mathbf{s}_{1,i} = \begin{bmatrix} \mathbf{s}_{1,1,i} \\ \mathbf{s}_{1,2,i} \end{bmatrix}$, $\mathbf{s}_{1,1,i} \in \mathbb{Z}_q^k, \mathbf{s}_{1,2,i} \in \mathbb{Z}_q^{\ell\cdot m}, \mathbf{s}_{2,i} \in \mathbb{Z}_q^{\ell\cdot m}, \mathbf{s}_{3,i} \in \mathbb{Z}_q^{k-n}$.
5. $Issuer_i$ send the credential share $cred_i$ that he generated to User.

- **AggreCred**$(\{cred_i\}_{i\in I})$: After collecting $|I|(|I| \geq t_I)$ credential shares, where t_I is dynamically adjusted by Trusted Authority based on the total number of issuers currently online. User aggregates these credential shares into a single credential $Cred$ through the following steps:

1. Parse $cred_i = \begin{bmatrix} \mathbf{s}_{1,i} \\ \mathbf{s}_{2,i} \\ \mathbf{s}_{3,i} \end{bmatrix}$, where $\mathbf{s}_{1,i} = \begin{bmatrix} \mathbf{s}_{1,1,i} \\ \mathbf{s}_{1,2,i} \end{bmatrix}$, $\mathbf{s}_{1,1,i} \in \mathbb{Z}_q^k, \mathbf{s}_{1,2,i} \in \mathbb{Z}_q^{\ell\cdot m}, \mathbf{s}_{2,i} \in \mathbb{Z}_q^{\ell\cdot m}, \mathbf{s}_{3,i} \in \mathbb{Z}_q^{k-n}$.
2. Let $j \in [m]$. If $j \in I$, parse $\mathbf{s}_{1,2,j} = [\mathbf{s}_{1,2,j,1} | \cdots | \mathbf{s}_{1,2,j,m}]^T \in \mathbb{Z}_q^{\ell\cdot m}$, otherwise $\mathbf{s}_{1,2,j} = \mathbf{0}^{\ell\cdot m}$. $\mathbf{s}$ is the vertical concatenation of $[\mathbf{s}_{1,2,j,j}]_{j\in[m]} \in \mathbb{Z}_q^{\ell\cdot m}$.
3. Compute $Cred = \begin{bmatrix} \sum_{i\in I} \mathbf{s}_{1,1,i} \\ \sum_{i\in I} \mathbf{s}_{1,2,i} \\ \sum_{i\in I} \mathbf{s}_{2,i} \\ \sum_{i\in I} \mathbf{s}_{3,i} \end{bmatrix} = \begin{bmatrix} \mathbf{s}_{1,1} \\ \mathbf{s}_{1,2} \\ \mathbf{s}_2 \\ \mathbf{s}_3 \end{bmatrix}$.
4. Store the indices of the valid credential issuers as the bit vector $\mathbf{b}$. If $\mathbf{b}[i] = 1$, it indicates that the $Issuer_i$ participated in the issuing process; otherwise, he did not.
5. User keeps $Cred$, $\mathbf{b}$, $\mathbf{s}$.

- **TransCred**$(Cred, \mathbf{b}, \{pk_i\}_{i\in[m]}, pdn)$: User wants to randomize the pseudonym before proving that he has a credential from issuers through the following steps:

1. Sample $\mathbf{R}' \xleftarrow{\$} \mathbb{Z}_q^{(k-n)\times(\ell\cdot m)}$ and compute $pdn' = pdn + \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{bmatrix} \cdot \mathbf{R}' = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{bmatrix} \cdot \mathbf{R} + \begin{bmatrix} \mathbf{0}^{n\times(\ell\cdot m)} \\ \mu \cdot \mathbf{G} \end{bmatrix} + \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{bmatrix} \cdot \mathbf{R}'$.
2. Compute $Cred' = \begin{bmatrix} \mathbf{s}_{1,1} \\ \mathbf{s}_{1,2} \\ \mathbf{s}_2 \\ \mathbf{s}_3 - \mathbf{R}' \cdot \mathbf{s}_2 \end{bmatrix}$.
3. Compute pk $= (\sum_{i\in I} \mathbf{A}_{0,i}, \mathbf{B}, \mathbf{u}) = (\mathbf{A}_0, \mathbf{B}, \mathbf{u})$.
4. Output $Cred'$, pdn', pk.

- **VerCred**$(Cred', \mathbf{b}, \{pk_i\}_{i\in[m]}, pk, pdn', \mathbf{s})$: Adjudication Committee checks the validity of the credential through following steps:

1. Recompute $\widehat{pk} = (\sum_{i\in[m],\mathbf{b}[i]=1} \mathbf{A}_{0,i}, \mathbf{B}, \mathbf{u})$ and check whether $\widehat{pk} = pk$.
2. Parse $pdn' = \begin{bmatrix} \mathbf{C}_1' \\ \mathbf{C}_2' \end{bmatrix}$ where $\mathbf{C}_1' \in \mathbb{Z}_q^{n\times(\ell\cdot m)}, \mathbf{C}_2' \in \mathbb{Z}_q^{\ell\times(\ell\cdot m)}$ and $Cred' = \begin{bmatrix} \mathbf{s}_{1,1} \\ \mathbf{s}_{1,2} \\ \mathbf{s}_2 \\ \mathbf{s}_3' \end{bmatrix}$ where $\mathbf{s}_{1,1} \in \mathbb{Z}_q^k, \mathbf{s}_{1,2} \in \mathbb{Z}_q^{\ell\cdot m}, \mathbf{s}_2 \in \mathbb{Z}_q^{\ell\cdot m}, \mathbf{s}_3' \in \mathbb{Z}_q^{k-n}$.
3. Check $\mathbf{D}\cdot\mathbf{s}_{1,1} + (\mathbf{A}_0 - |I| \cdot \mathbf{G})\cdot\mathbf{s} + \mathbf{G}\cdot\mathbf{s}_{1,2} + (\mathbf{B} + \mathbf{C}_2')\cdot\mathbf{s}_2 + \mathbf{A}_2\cdot\mathbf{s}_3' = |I| \cdot \mathbf{u} \ mod\ q$.
4. If all statements and equations hold, Adjudication Committee accepts the credential and computes $\sigma = H(Cred')$. Then, update the set of accumulator $X' = X \cup \{\sigma\}$.

5. Construct the vector $\mathbf{z} = [\mathbf{z}_1^T | \cdots | \mathbf{z}_M^T]^T$, where each $\mathbf{z}_i$ is a c-dimensional nonzero vector with all terms are 1 if $i \in X'$, otherwise, $\mathbf{z}_i = \mathbf{0}^c$. Sample $\mathbf{r} \leftarrow D_{\mathbb{Z}_q^b,\alpha}$. Then, update the accumulator value $\mathbf{v} = \mathbf{U}\mathbf{z} + \mathbf{A}'\mathbf{r}$.
6. Compute the witness for σ

$$\mathbf{w} = \sum_{j \in X', j \neq \sigma} \mathbf{S}_j \mathbf{z}_j + \mathbf{r}$$

where $\mathbf{z}_j$ is calculated from X'.
7. Output $\mathbf{v}$,$\mathbf{w}$ for σ corresponding to $Cred'$.

- **AggreWit**($\{\mathbf{w}_i\}_{i \in X}$): If User has a set of valid membership witnesses, he aggregates the membership witnesses through the following steps:

1. Suppose the number of X is N. Then, compute $\mathbf{w}' = \sum_{i \in X} \mathbf{w}_i$.
2. Output the aggregated membership witness $\mathbf{w}'$.

- **BatchVer**($\mathbf{w}', \mathbf{v}, pp, X$): The User wants to prove that he has a set of credentials. If User owns a set of valid credentials, given aggregated membership witnesses $\mathbf{w}'$ and the accumulator value $\mathbf{v}$, Verifier validates their correctness through the following steps:

1. Let $\mathbf{y} = [\mathbf{y}_1^T | ... | \mathbf{y}_i^T | ... | \mathbf{y}_M^T]^T$, $\mathbf{y}_i$ is an c-dimensional nonzero vector that all terms are 1 if $i \in X$, otherwise, $\mathbf{y}_i = \mathbf{0}^c$. Check $N \cdot \mathbf{v} = \mathbf{A}'\mathbf{w}' + \mathbf{U} \cdot \mathbf{y}$.
2. Check $\|\mathbf{w}'\|_\infty \leq \gamma$ or $\|\mathbf{w}'\| \leqslant \gamma_1$.
3. Only when the above statements have been verified as correct can the verifier accept them.

4. Analysis

4.1. Correctness

We analyze the correctness of our scheme from two aspects, including credential correctness verification and membership witness correctness verification.

1. credential correctness: The credential share $cred_i$ issued by $Issuer_i$ meets:

$$\mathbf{D} \cdot \mathbf{s}_{1,1,i} + \mathbf{A}_{0,i} \cdot \mathbf{s}_{1,2,i} + (\mathbf{B} + \mathbf{C}_2) \cdot \mathbf{s}_{2,i} + \mathbf{A}_2 \cdot \mathbf{s}_{3,i} = \mathbf{u} \ mod \ q.$$

Suppose that the first t issuers have issued credential shares:

$$\mathbf{D} \cdot \mathbf{s}_{1,1,1} + \mathbf{D} \cdot \mathbf{T}_1' \cdot \mathbf{s}_{1,2,1} + \mathbf{G} \cdot \mathbf{s}_{1,2,1} + (\mathbf{B} + \mathbf{C}_2) \cdot \mathbf{s}_{2,1} + \mathbf{A}_2 \cdot \mathbf{s}_{3,1} = \mathbf{u} \ mod \ q,$$

...

$$\mathbf{D} \cdot \mathbf{s}_{1,1,t} + \mathbf{D} \cdot \mathbf{T}_t' \cdot \mathbf{s}_{1,2,t} + \mathbf{G} \cdot \mathbf{s}_{1,2,t} + (\mathbf{B} + \mathbf{C}_2) \cdot \mathbf{s}_{2,t} + \mathbf{A}_2 \cdot \mathbf{s}_{3,t} = \mathbf{u} \ mod \ q.$$

Add up the left and right sides of the above equation separately:

$$\begin{aligned} &\mathbf{D} \cdot \sum_{i \in [t]} \mathbf{s}_{1,1,i} + (\mathbf{A}_0 - |I| \cdot \mathbf{G}) \cdot \mathbf{s} + \mathbf{G} \cdot \sum_{i \in [t]} \mathbf{s}_{1,2,i} \\ &+ (\mathbf{B} + \mathbf{C}_2) \cdot \sum_{i \in [t]} \mathbf{s}_{2,i} + \mathbf{A}_2 \cdot \sum_{i \in [t]} \mathbf{s}_{3,i} \\ &= |I| \cdot \mathbf{u} \ mod \ q. \end{aligned}$$

After credential transformation,we can obtain the following equation:

$$\mathbf{A}_2 \cdot \mathbf{s}_3' = |I| \cdot \mathbf{u} \ mod \ q - [\mathbf{D} \cdot \mathbf{s}_{1,1} + (\mathbf{A}_0 - |I| \cdot \mathbf{G}) \cdot \mathbf{s} + \mathbf{G} \cdot \mathbf{s}_{1,2} + (\mathbf{B} + \mathbf{C}_2) \cdot \mathbf{s}_2 + \mathbf{A}_2 \cdot \mathbf{R}' \cdot \mathbf{s}_2].$$

If it satisfies $\mathbf{s}_3' = \mathbf{s}_3 - \mathbf{R}' \cdot \mathbf{s}_2$, the above equation holds.It can be seen that the credential verification of our scheme is correct.
2. membership witness correctness: It is known that there are N credentials, the witness for $i \in [N]$ is:

$$\mathbf{w}_i = \sum_{j \in [N], j \neq i} \mathbf{S}_j \mathbf{z}_j + \mathbf{r}.$$

The aggregated membership witness is:

$$\mathbf{w}' = (N-1) \sum_{j \in [N]} \mathbf{S}_j \mathbf{z}_j + N\mathbf{r}.$$

According to **BatchVer**, it can be obtained that the accumulator value is:

$$\begin{aligned} \mathbf{v} &= \frac{\mathbf{A}'\mathbf{w}' + \mathbf{U} \cdot \mathbf{y}}{N} \\ &= \frac{\mathbf{A}'\mathbf{w}' + \sum_{i \in [N]} \mathbf{U}_i \cdot \mathbf{y}_i}{N} \\ &= \frac{\mathbf{A}'[(N-1)\sum_{j \in [N]} \mathbf{S}_j \mathbf{z}_j + N\mathbf{r}] + \sum_{i \in [N]} \mathbf{U}_i \cdot \mathbf{y}_i}{N} \\ &= \frac{N \sum_{i \in [N]} \mathbf{U}_i \cdot \mathbf{y}_i + N\mathbf{A}'\mathbf{r}}{N} \\ &= \sum_{i \in [N]} \mathbf{U}_i \mathbf{y}_i + \mathbf{A}'\mathbf{r}. \end{aligned}$$

If it satisfies $\|\mathbf{w}'\|_\infty \leq O(\alpha\sqrt{cM})$ or $\|\mathbf{w}'\| \leq O(\alpha(b + cM))$ and $\mathbf{v} = \sum_{i \in [N]} \mathbf{U}_i \mathbf{y}_i + \mathbf{A}'\mathbf{r}$, the above equation holds.It can be seen that the membership witness verification of our scheme is correct.

4.2. Security analysis

This part,we evaluate the security of our scheme by performing the following games between a challenger C and a Probabilistic Polynomial Time adversary $\mathcal{A}$, ensuring that it meets security requirements such as unforgeability, blindness, and unlinkability.A simulator S is utilized to compromise the security of the scheme established by C.

Theorem 4.1. *Our scheme achieves unforgeability if the LWE and SIS assumptions hold.*

Proof. Assume that $\mathcal{A}$ can compromise the EU-CMA security of our scheme with a non-negligible advantage of ϵ. S is typically believed to exploit the ability of C to forge credentials. The specific process is as follows:

1. *Setup Phase:* C generates the public system parameters pp and sends Ask to Adjudication Committee. After that, S runs **KeyGen** to generate sk_i and corresponding pk_i for $Issuer_i$ by utilizing pp. S sends pp and $\{pk_i\}_{i \in [m]}$ to $\mathcal{A}$.
2. *Query Phase:* $\mathcal{A}$ submits adaptive queries to the oracles $\{\mathcal{O}^{PdnGen}, \mathcal{O}^{Issue}, \mathcal{O}^{AggreCred}\}$.
$\mathcal{O}^{PdnGen}$: $\mathcal{A}$ queries the pseudonym generation oracles with the message μ, C runs **PdnGen** to get the pseudonym pdn, then sends pdn to $\mathcal{A}$ through S.
$\mathcal{O}^{Issue}$: $\mathcal{A}$ queries the credential share generation oracles with the pseudonym pdn and the corresponding Non-Interactive Zero-Knowledge Proof Θ.C computes credential share $cred_i$ to $\mathcal{A}$.
$\mathcal{O}^{AggreCred}$: $\mathcal{A}$ queries the credential aggregation oracle with the credential shares $\{cred_i\}_{i \in I}$, S returns the single full credential $Cred$ to $\mathcal{A}$.
3. *Forgery:* $\mathcal{A}$ generates the forged credential $Cred^*$ connected with μ^*. If the following conditions are met, the forgery is considered valid.

(1) : $pdn^* = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{bmatrix} \cdot \mathbf{R} + \begin{bmatrix} \mathbf{0}^{n \times (\ell \cdot m)} \\ \mu^* \cdot \mathbf{G} \end{bmatrix} \in \mathbb{Z}_q^{(n+\ell) \times (\ell \cdot m)} = \begin{bmatrix} \mathbf{C}_1 \\ \mathbf{C}_2^* \end{bmatrix}$;
(2) : $\mathbf{D} \cdot \mathbf{s}_{1,1,i}^* + \mathbf{A}_{0,i} \cdot \mathbf{s}_{1,2,i}^* + (\mathbf{B} + \mathbf{C}_2^*) \cdot \mathbf{s}_{2,i}^* + \mathbf{A}_2 \cdot \mathbf{s}_{3,i}^* = \mathbf{u} \bmod q$;
(3) : Θ is vaild;
(4) : $\mu^* \neq \mu$.

4. *Remark:* In the simulation described, there are two scenarios where $\mathcal{A}$ tries to create an invalid proof for the combined credential $Cred^*$ in an effort to deceive the verification process.

(1) $\mathcal{A}$ does not possess a valid pseudonym, yet manages to successfully forge a proof: In the pseudonym generation algorithm, User creates a pseudonym and then generates a proof for it. If User produces a valid proof and obtains a credential share, which means the proof has passed verification. This would contradict the soundness property of the NIZK described in 2.5;

(2) Despite possessing fewer than t_I credential shares, $\mathcal{A}$ manages to construct a legitimate aggregated credential:From the specific construction of the algorithm in 3.4, we understand that the aggregation of credentials requires the addition of certain parts from at least t credential shares. Moreover, this process not only outputs the aggregated credential but also the combined public key of the valid issuers. Therefore, to pass the verification, $\mathcal{A}$ must forge $(\mathbf{A}^*_{0,i}, \mathbf{B}, \mathbf{u})$ such that $pk = (\sum_{i\in I} \mathbf{A}_{0,i}, \mathbf{B}, \mathbf{u}) + (\mathbf{A}^*_{0,i}, \mathbf{B}, \mathbf{u}) = (\mathbf{A}_0, \mathbf{B}, \mathbf{u})$ and $\mathbf{D}\cdot\mathbf{s}^*_{1,1,i} + \mathbf{A}_{0,i}\cdot\mathbf{s}^*_{1,2,i} + (\mathbf{B}+\mathbf{C}^*_2)\cdot\mathbf{s}^*_{2,i} + \mathbf{A}_2\cdot\mathbf{s}^*_{3,i} = \mathbf{u} \bmod q$, which would allow $\mathcal{A}$ to obtain a forged credential share through $(\mathbf{A}^*_{0,i}, \mathbf{B}, \mathbf{u})$. However, this would compromise the LWE problem, hence $\mathcal{A}$ cannot possibly pass the verification. The probability of the adversary successfully forging a credential is $\frac{(\epsilon')^t}{q}$,where t is the number of forged shares, and ϵ' represents the probability of successful forgery. Therefore, the advantage $\mathrm{Adv}^{\mathrm{EU-CMA}}_{\mathcal{A}} = \frac{(\epsilon')^t}{q}$ is negligible.

Theorem 4.2. *Our scheme achieves blindness if the LWE assumption holds.*

Proof. Assume that $\mathcal{A}$ can win the security game Exp^{Blind} with a non-negligible advantage of ϵ. $\mathcal{S}$ is typically believed to exploit C's ability to break the blindness of our scheme. The specific process is as follows:

1. *Setup Phase:* C generates the public system parameters pp. After that, $\mathcal{S}$ runs **KeyGen** to generate pk_i for $Issuer_i$ by utilizing pp and sends Ask to Adjudication Committee. $\mathcal{S}$ queries the pseudonym generation oracle $\mathcal{O}^{PdnGen}$ to get two pseudonyms (pdn_0, pdn_1) on message (μ_0, μ_1). Then, $\mathcal{S}$ sends $(pp, pdn_0, pdn_1, \{pk_i\}_{i\in[m]})$ to $\mathcal{A}$.
2. *Query Phase:* $\mathcal{A}$ submits adaptive queries to the oracles $\{\mathcal{O}^{Issue}, \mathcal{O}^{Blind}\}$.
   $\mathcal{O}^{Issue}$: $\mathcal{A}$ queries the pseudonym generation oracle $\mathcal{O}^{PdnGen}$ with the message μ and gets a corresponding pseudonym pdn. Then $\mathcal{S}$ is capable of emulating the corresponding Non-Interactive Zero-Knowledge Proof Θ and continues with subsequent operations as usual.
   $\mathcal{O}^{Blind}$: $\mathcal{A}$ sends μ_b to $\mathcal{O}^{PdnGen}$ and receives pdn_b,where $b \in \{0,1\}$. $\mathcal{S}$ then simulates the corresponding proof Θ_b and carries out the interaction with Issuers to acquire credential shares.After gathering t partial credentials, $\mathcal{S}$ consolidates the credentials by inquiring the aggregate credential oracle $\mathcal{O}^{AggreCred}$ to obtain $Cred_b$.$\mathcal{S}$ subsequently sends $Cred_b$ to $\mathcal{A}$.
3. *Guess:* $\mathcal{A}$ sends a binary digit b^* as a guess for $Cred_b$ through $\mathcal{S}$ to C. C determines whether $b^* = b$. If they are equal,it means that $\mathcal{A}$ wins the security game Exp^{Blind}.
4. *Remark:* In the simulation described, $\mathcal{A}$ wins the game mainly due to a key property. The commitment scheme used in the pseudonym generation process provides the hiding property, a feature that is based on the LWE assumption. $\mathcal{S}$ uses the forgeries attempted by $\mathcal{A}$ to generate a valid forgery under the LWE assumption. If $\mathcal{A}$ can win the blindness game with a non-negligible advantage, then $\mathcal{S}$ can also solve the LWE problem with the same advantage. This indicates that if the LWE assumption hold, it is impossible for $\mathcal{A}$ to generate a valid credential without knowing the content of the message. In summary, the advantage $\mathrm{Adv}^{\mathrm{Blind}}_{\mathcal{A}} = \epsilon_{LWE}$ is negligible.

Theorem 4.3. *Our scheme achieves unlinkability if the LWE assumption holds.*

Proof. Assume that $\mathcal{A}$ can win the security game Exp^{Unlink} with a non-negligible advantage of ϵ. $\mathcal{S}$ is typically believed to exploit C's ability to break the unlinkability of our scheme. The specific process is as follows:

1. *Setup Phase:* C generates the public system parameters pp and sends Ask to Adjudication Committee.After that, $\mathcal{S}$ utilizes pp to generate pk_i for $Issuer_i$.$\mathcal{S}$ computes two aggregated membership witnesses $(\mathbf{w}'_0, \mathbf{w}'_1)$ for two sets $(\mathbf{w}_1, \ldots, \mathbf{w}_N)_0$, $(\mathbf{w}_1, \ldots, \mathbf{w}_N)_1$. Then, $\mathcal{S}$ sends $(pp, \mathbf{w}'_0, \mathbf{w}'_1, \{pk_i\}_{i\in[m]})$ to $\mathcal{A}$.
2. *Query Phase:* $\mathcal{A}$ submits adaptive queries to the oracles $\{\mathcal{O}^{Issue}, \mathcal{O}^{Unlink}\}$.
   $\mathcal{O}^{Issue}$: $\mathcal{A}$ queries the pseudonym generation oracle $\mathcal{O}^{PdnGen}$ with message μ and gets a corresponding pseudonym pdn. Then $\mathcal{S}$ is capable of emulating the corresponding Non-Interactive Zero-Knowledge Proof Θ and continues with the subsequent operations as usual.
   $\mathcal{O}^{Unlink}$: $\mathcal{S}$ selects two sets of witness $\{(\mathbf{w}_1, \ldots, \mathbf{w}_N)_0, (\mathbf{w}_1, \ldots, \mathbf{w}_N)_1\}_0$, $\{(\mathbf{w}_1, \ldots, \mathbf{w}_N)_0, (\mathbf{w}_1, \ldots, \mathbf{w}_N)_1\}_1$. $\mathcal{S}$ carries out the algorithm **AggreWit** to obtain the corresponding aggregate witnesses $(\mathbf{w}'_0, \mathbf{w}'_1)_0$ and $(\mathbf{w}'_0, \mathbf{w}'_1)_1$, where the first set implies that they are from the same user, and the second set implies that they are from different user. $\mathcal{S}$ flips a coin b which can be either 0 or 1 and sends $(\mathbf{w}'_0, \mathbf{w}'_1)_b$ to $\mathcal{A}$.
3. *Guess:* $\mathcal{A}$ sends a binary digit b^* as a guess for the set of aggregate witnesses through $\mathcal{S}$ to C. C determines whether $b^* = b$. If they are equal,it means that $\mathcal{A}$ wins the security game Exp^{Unlink}.
4. *Remark:* In the simulation described, $\mathcal{A}$ wins the game mainly due to a key factor. The method provided in [39] is used for aggregating proof, which conceals the proof to be aggregated in a zero-knowledge form.This implies that during the **AggreWit**, no additional information is disclosed beyond the validity of the aggregated witness, and it also ensures the unlinkability between different batch verification processes. The security of aggregating witness relies on the LWE problem. This is because that there is a correspondence between computational assumption and the unlinkability guarantee. In the part of membership witness correctness, we have come to a conclusion that the aggregated $\mathbf{w}' = (N-1)\sum_{j\in[N]}\mathbf{S}_j\mathbf{z}_j + N\mathbf{r}$. If $\mathcal{A}$ can win the unlinkability game with a non-negligible advantage, then $\mathcal{S}$ can also solve the LWE problem with the same advantage. This indicates that if the LWE assumption holds, it is impossible for $\mathcal{A}$ to link the witness without knowing the content of the message. In summary, the advantage $\mathrm{Adv}^{\mathrm{Unlink}}_{\mathcal{A}} = \epsilon_{LWE}$ is negligible.

5. Performance comparison

In this section, we provide our evaluation results which are derived from comparing the proposed scheme with several well-known schemes,including Coconut [21],SFY+ [25] and DTACB [41]. To more clearly present the results, our analysis focuses on the key functionalities:

- *Threshold Issuance*: Threshold issuance provides a decentralized method for issuing credentials. In traditional systems, the issuance of credentials typically relies on a central issuer, making the system susceptible to single point of failure. Requiring multiple issuers to participate in issuance process reduces the risk.
- *Blind Issuance*: Blind issuance can prevent issuers from learning the private information of the credential applicant, ensure the blindness of credentials, and allow users to use credentials multiple times while maintaining anonymity.

Table 2
Comparison of functionality.

Scheme	Threshold issuance	Blind issuance	Dynamic threshold	Batch verification	Post-quantum security
SPS-EQ [40]	✗	✓	✗	✗	✗
Coconut [21]	✓	✓	✗	✗	✗
SFY+ [25]	✓	✓	✗	✗	✗
DTACB [41]	✓	✓	✓	✓	✗
Ours	✓	✓	✓	✓	✓

Table 3
Computational complexity.

Scheme	KeyGen	PdnGen	Issue	AggreCred
Coconut [21]	$2IT_{E_{\mathbb{G}_2}}$	$6T_{E_{\mathbb{G}_1}}+T_{h_1}$	$4T_{E_{\mathbb{G}_1}}+T_{h_1}$	$\lvert X\rvert T_{E_{\mathbb{G}_1}}+2\lvert X\rvert T_{A_{\mathbb{Z}_q^*}}+2\lvert X\rvert T_{M_{\mathbb{Z}_q}}$
SFY+ [25]	$3I(T_{E_{\mathbb{G}_1}}+T_{E_{\mathbb{G}_2}})$	$(t+3)T_{E_{\mathbb{G}_1}}+2IT_{E_{\mathbb{G}_2}}+(t-1)(T_{A_{\mathbb{Z}_q^*}}+T_{M_{\mathbb{Z}_q^*}})+2T_P+T_{h_1}$	$5T_{E_{\mathbb{G}_1}}+2T_P+T_{E_{\mathbb{G}_2}}+T_{M_{\mathbb{G}_2}}+T_{h_1}$	$2\lvert X\rvert T_{M_{\mathbb{Z}_q^*}}+\lvert X\rvert T_{A_{\mathbb{Z}_q^*}}+\lvert X\rvert T_{E_{\mathbb{G}_1}}+2T_P+2T_{E_{\mathbb{G}_2}}+T_{M_{\mathbb{G}_2}}$
DTACB [41]	$IT_{E_{\mathbb{G}_2}}$	$12T_{E_{\mathbb{G}_1}}+T_{h_1}$	$3T_{E_{\mathbb{G}_1}}+T_{E_{\mathbb{G}_2}}+T_{M_{\mathbb{G}_1}}+T_{h_1}$	$\lvert X\rvert T_{M_{\mathbb{G}_1}}$
Ours	$I\ell^2 m(T_{A_{\mathbb{Z}_q}}+T_{M_{\mathbb{Z}_q}})$	$m\ell(n+\ell)(2T_{M_{\mathbb{Z}_q}}+T_{A_{\mathbb{Z}_q}})$	$T_{s,p}$	$\lvert X\rvert\ell(2k+2\ell m-n)T_{A_{\mathbb{Z}_q}}$
Scheme	**TransCred**	**VerCred**	**AggreWit**	**BatchVer**
Coconut [21]	$3T_{E_{\mathbb{G}_1}}+2T_{E_{\mathbb{G}_2}}$	$2T_{E_{\mathbb{G}_2}}+T_{E_{\mathbb{G}_1}}+T_{M_{\mathbb{G}_1}}+T_P$	N/A	N/A
SFY+ [25]	$6T_{E_{\mathbb{G}_1}}+2T_{E_{\mathbb{G}_2}}+T_{h_2}$	$T_{E_{\mathbb{G}_1}}+T_{E_{\mathbb{G}_2}}+7T_P+T_{M_{\mathbb{G}_2}}+T_{h_2}$	N/A	N/A
DTACB [41]	$2T_{E_{\mathbb{G}_1}}+2\lvert X\rvert T_{E_{\mathbb{G}_2}}$	$(\lvert X\rvert+2)T_{E_{\mathbb{G}_2}}+3T_{E_{\mathbb{G}_1}}+MT_{M_{\mathbb{Z}_q}}+2T_P+T_{h_2}$	$\lvert P\rvert T_{M_{\mathbb{G}_1}}+\lvert P\rvert T_{E_{\mathbb{G}_1}}$	$16T_{E_{\mathbb{G}_1}}+6T_{E_{\mathbb{G}_2}}+7T_{E_{\mathbb{G}_T}}+15T_P+(\lvert P\rvert+12)T_{M_{\mathbb{G}_2}}+5T_{A_{\mathbb{G}_2}}+T_{h_2}$
Ours	$2[(n+\ell)m\ell+k-n](T_{M_{\mathbb{Z}_q}}+T_{A_{\mathbb{Z}_q}})$	$(2a+b+l)T_{M_{\mathbb{Z}_q}}+(a+b)T_{A_{\mathbb{Z}_q}}+T_H$	$\lvert P\rvert bT_{A_{\mathbb{Z}_q}}$	$3aT_{M_{\mathbb{G}_q}}+aT_{A_{\mathbb{G}_q}}$

Note: The notations used in computational complexity analysis are: I is the number of issuers, t is the threshold, $\lvert X\rvert$ is the number of credential shares, $\lvert P\rvert$ is the number of user's valid credentials. $T_{E_{\mathbb{G}_1}}$ is the exponentiation time in $\mathbb{G}_1$, $T_{E_{\mathbb{G}_2}}$ is the exponentiation time in $\mathbb{G}_2$, $T_{E_{\mathbb{G}_T}}$ is the exponentiation time in $\mathbb{G}_T$.$T_{M_{\mathbb{G}_1}}$ is the multiplication operation time in $\mathbb{G}_1$, $T_{M_{\mathbb{G}_2}}$ is the multiplication operation time in $\mathbb{G}_2$, T_P is the pairing operation time. $T_{A_{\mathbb{Z}_q}}$ is the addition operation time in $\mathbb{Z}_q^*$, $T_{M_{\mathbb{G}_q}}$ is the multiplication operation time in $\mathbb{Z}_q^*$.T_H is the hash computation time of $H:\mathbb{Z}_q^{2k+2\ell m-n}\to\mathbb{Z}_q^*$.$T_{h_1}$ is the hash computation time of $H_1:\{0,1\}^*\to\mathbb{G}_1$.$T_{h_2}$ is the hash computation time of $H_2:\{0,1\}^*\to\mathbb{Z}_q^*$.$T_{s,p}$ is the running cost of algorithm SamplePre.

Table 4
Storage requirements and communication overhead.

Scheme	Storage requirements		Communication overhead	
	Public key size	Credential size	Credentials issuance	Credentials verification
Coconut [21]	$3I\lvert\mathbb{G}_2\rvert$	$\lvert\mathbb{G}_2\rvert+\lvert\mathbb{G}_1\rvert$	$t\lvert\mathbb{G}_2\rvert+t\lvert\mathbb{G}_1\rvert$	$6N\lvert\mathbb{G}_1\rvert$
SFY+ [25]	$6I\lvert\mathbb{G}_1\rvert+6I\lvert\mathbb{G}_2\rvert$	$2\lvert\mathbb{G}_1\rvert$	$2t\lvert\mathbb{G}_1\rvert$	$5N\lvert\mathbb{G}_1\rvert+N\lvert\mathbb{G}_2\rvert$
DTACB [41]	$I\lvert\mathbb{G}_2\rvert$	$2\lvert\mathbb{G}_1\rvert$	$2t\lvert\mathbb{G}_1\rvert+t\lvert\mathbb{G}_2\rvert$	$3\lvert\mathbb{G}_1\rvert+4\lvert\mathbb{G}_2\rvert+\lvert\mathbb{G}_T\rvert+5\log q$
Ours	$a(cM+b)\log q$	$(2k+2\ell m-n)\log q$	$t(2k+2\ell m-n)\log q$	$(a+b)\log q$

Note: The notations used in storage requirements and communication overhead analysis are: I is the number of issuers, t is the threshold, N is the number of user's valid credentials. $\lvert\mathbb{G}_1\rvert$ is the element sizes in group $\mathbb{G}_1$.$\lvert\mathbb{G}_2\rvert$ is the element sizes in group $\mathbb{G}_2$.$\lvert\mathbb{G}_T\rvert$ is the element sizes in group $\mathbb{G}_T$.

- *Dynamic Threshold*: Dynamic threshold allows for the adjustment of threshold in different scenarios, making the system more flexible. For example, in some cases, more participants may be required to reach a consensus, while in other cases, fewer participants may be needed.
- *Batch Verification*: Batch verification can validate a user's multiple credentials in one go, significantly improving verification efficiency compared to validating each credential individually. In addition, unlike traditional batch processing algorithms, the algorithm proposed in this paper allows users to hide their credential set.
- *Post-quantum Security*: Previous schemes are mostly based on number theory, ultimately relying on the hardness of integer factorization or the discrete logarithm problem in certain groups. Shor's algorithm can solve these problems, which means that previous schemes may not maintain their security when facing quantum attacks. In contrast, there are no effective quantum algorithms for the problems commonly used in lattice cryptography, which is considered to be resistant to quantum attacks.

Table 2 intuitively presents a comparison of functionality between the scheme proposed in this paper and previous relevant schemes.Our scheme stands out among the relevant schemes because it combines all functionalities which contains threshold issuance,blind issuance, dynamic threshold,batch verification and post-quantum security.While, SPS-EQ scheme [40],Coconut scheme [21], SFY+ scheme [25] and DTACB [41] are vulnerable to quantum attacks due to the need for certain pairing and exponentiation operations. The SPS-EQ scheme [40] only supports blind issuance. Compared to it, the Coconut scheme [21] and SFY+ scheme [25] also support threshold issuance. Finally, DTACB [41] offers relatively comprehensive functionalities which contains threshold issuance,blind issuance,dynamic threshold and batch verification.

Tables 3 and 4 offer a theoretical analysis of the schemes proposed in [21,25,41] and ours. Table 3 intuitively shows the comparison of computational complexity. Table 4 provides a detailed comparison in terms of storage requirements and communication overhead. Since the schemes proposed in [21,25,41] are all built on the foundation of bilinear pairing, the principle of calculation is utilizing certain pairing and exponentiation operations on group generator elements to complete aggregation and verification [42]. Ours employs matrix operations to complete the corresponding algorithms.

As shown in Tables 3 and 4, the batch verification algorithm achieves constant computational complexity and communication overhead regardless of credential quantity. Even if the scale of credentials grows exponentially, the verification process can still maintain constant efficiency. This feature originates from the design of membership witnesses aggregation operations for multiple credentials, compressing

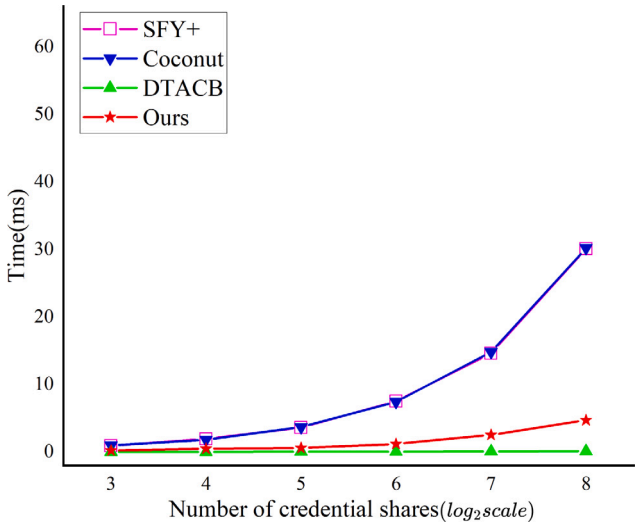


Fig. 5. Credential shares aggregation algorithm.

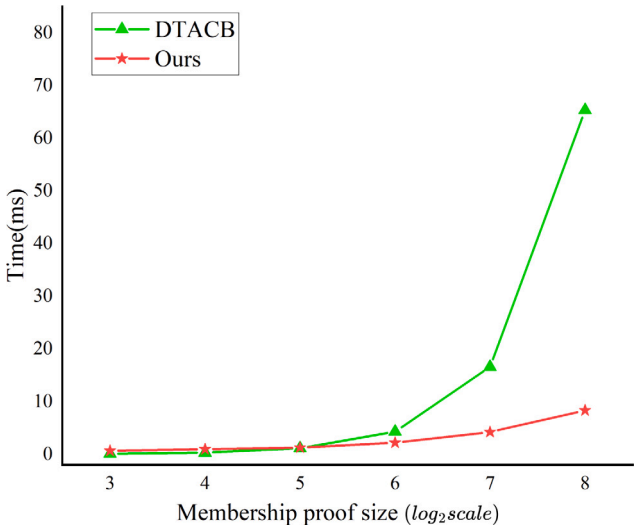


Fig. 6. Credential membership witnesses aggregation algorithm.

computational complexity to O (1). In the credential issuance process, as each issuer needs to independently generate and transmit credential shares, the computation time and bandwidth consumption are linearly correlated with the number of issuers. This asymmetric feature makes this scheme particularly suitable for multi-issuer collaboration scenarios: when the number of issuers is controllable, the system can leverage the constant cost advantage of batch verification to support efficient verification of massive credentials, demonstrating significant scalability advantages in large-scale credential application scenarios such as electronic voting and federated identity management.

In order to achieve a more accurate comparison, we utilize Python 3.11 to conduct simulation experiments. The experiments are performed on a laptop with an Intel(R) Core(TM) i5-11320H CPU @ 3.20 GHz, 16 GB RAM, running the Windows 11 64-bit operating system. After 1000 executions, we determine the average running time of key algorithms and compare it with the previous system. Fig. 5 is the performing time of credential shares aggregation algorithm. Fig. 6 is the performing time of credential membership witnesses aggregation algorithm. Fig. 7 is the performing time of credentials batch verification algorithm.

Fig. 5 intuitively shows the comparison of credential shares aggregation algorithm execution time in ours, DTACB, SFY+ and Coconut with credential share numbers ranging from 2^3 to 2^8. It can be seen that the execution time in ours is significantly less than SFY+ and Coconut but slightly greater than DTACB. SFY+ and Coconut calculate Lagrangian coefficients for each credential share, then perform exponentiation in $\mathbb{G}_1$, which has a high computational complexity. DTACB and our scheme only perform addition operations in $\mathbb{G}_1$ and $\mathbb{Z}_q$, respectively. However, the time cost of addition operation in $\mathbb{Z}_q$ is naturally higher than $\mathbb{G}_1$. It is worth noting that even with a sharp increase in the number of credential shares, the time cost is still much lower than that of exponentiation.

Due to the absence of credential membership witnesses aggregation algorithm in SFY+ and Coconut, we only compare ours with DTACB. The results are displayed in Fig. 6 with credential membership witness size spanning from 2^3 to 2^8. It is apparent that ours has a slightly higher time cost than DTACB before the quantity reaches 2^5, but it is much lower afterwards. DTACB carries out accumulator polynomials multiplication for the witness set, then performs Partial fraction decomposition (PFD) and finally executes exponentiation in $\mathbb{G}_1$. Ours only requires the addition operation in $\mathbb{Z}_q$ for witness set. Although matrix operation may result in slightly higher time cost compared to DTACB when the number of witnesses is small, this increase will be relatively slower as the number increases.

Fig. 7 intuitively reveals the comparison of credentials verification algorithm execution time in ours, DTACB, SFY+ and Coconut with credential numbers ranging from 2^3 to 2^7. It can be seen that the execution time in ours is less than DTACB, SFY+ and Coconut. SFY+ and Coconut do not support credentials batch verification resulting in significantly higher time costs. As exhibited in Table 3, DTACB performs exponentiation and pairing operations of group generator elements, as well as addition and multiplication operations of integers, while our scheme only performs constant addition and multiplication operations. Therefore, the execution time in ours is less than DTACB.

The comparative analysis presented above demonstrates that our system contains more comprehensive functionalities and exhibits superior performance when compared with the previous systems proposed in [21,25,41].

6. Conclusion

In this paper,we propose the first lattice-based anonymous credential system which is resistant to quantum attacks in a decentralized environment. Our system allows issuers to dynamically change while enabling threshold adjustments without rewinding. By using a cryptographic accumulator,it supports batch verification of multiple credentials. We have demonstrated that this system satisfies unforgeability, blindness and unlinkability. Compared with other anonymous credential schemes, our system has a distinct performance advantage. It provides a theoretical framework for constructing lattice-based decentralized anonymous credentials, which contributes to promoting future research in this area.Taking inspiration from the service centric attribute-based block level two-step encryption scheme proposed in [43], we will further explore the generation and management of service level keys matched by user attributes to generate credentials in cloud environments, in order to enhance cloud security.

CRediT authorship contribution statement

Yihua Zhou: Writing – review & editing, Validation. **Shumiao Liu:** Writing – original draft, Writing – review & editing, Validation. **Yuguang Yang:** Funding acquisition. **Weimin Shi:** Software. **Zhenhu Ning:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

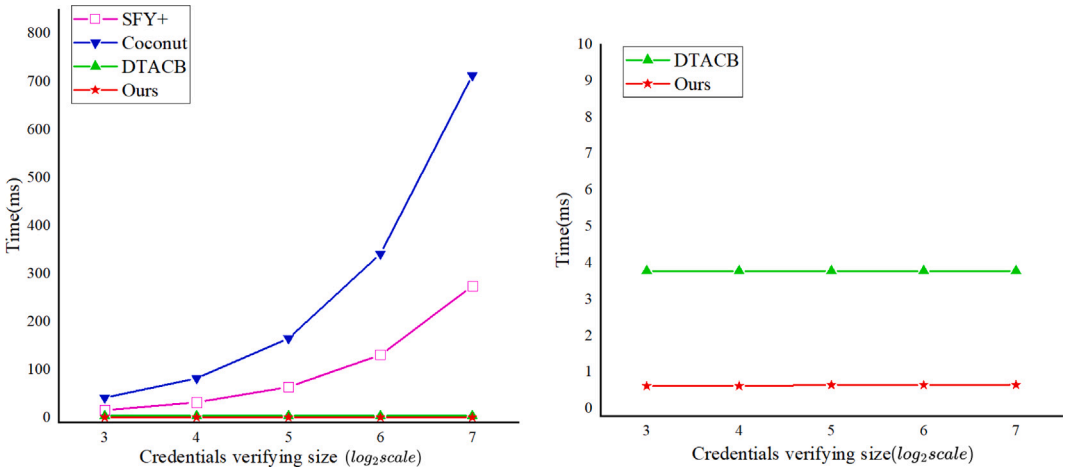


Fig. 7. Credentials batch verification algorithm.

Data availability

Data will be made available on request.

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